ESTR1002 Fall 2025

Lab Exercise #8

Due: 19 Nov 2025

Separate your answer for each question in a different .c file. Compress all files into a .zip and submit it to Blackboard.

Question 1

We want to climb up the stairs. Every time we can choose to move up 1 or 2 steps.
If there are only 3 steps in total, we can have 3 ways to climb up the stairs (111, 12, 21).
How about we have more steps? How many ways to climb up the stairs with totally *N* steps?

Input: integer $1 \leq N \leq 40$.
Output: the number of ways to climb up the stairs.

Question 2

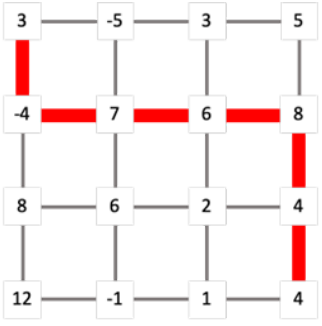
We have a 10 x 10 chessboard. A chess piece is put at the most bottom-left square of the chessboard. Every time we can move the chess to 1) any square on its right, or 2) any square on its top.

Input: none
Output: the number of ways to go to the most top-right square of the chessboard.

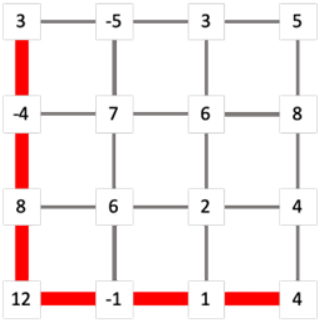
You need to compute the answer via a program (not just printing the final answer you computed somewhere else).

Question 3

Consider a N x N lattice. Every corner point of the lattice has an integral score. We start from the upper-left corner of the lattice and end at the lower-right corner of the lattice. We can only walk downward or rightward. Our goal is to collect the maximum score sum along the path.
For example, the images below illustrate two possible paths in a 3 x 3 lattice that collect different scores along their paths.



Path1 (total score: 28)



Path2 (total score: 23)

Input: the input contains two parts. The first line is an integer n (0 < n <= 10), which indicates the size of the lattice. Each of the subsequent n+1 lines contains n+1 integers (separated by spaces), forming an (n+1)x(n+1) matrix of the scores at the corner points of the lattice.
Output: the maximum score that you can obtain by drawing a path from the upper-left corner to the lower-right corner, with right and down movements only.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
1
0 1
2 0
2
```

Sample Run #2:

```
2
2 -3 5
-9 0 4
93 1 3
90
```

Question 4

Suppose we have a 4 x 4 matrix. Each entry is either 0 or 1. We choose one column and one row at every move. The corresponding entries in the chosen row and column will be flipped (0 to 1, 1 to 0).

The following is an example. The shaded row and column are chosen.

0	1	1	1
1	0	0	1
0	1	1	0
1	0	0	0

After the flip, it becomes

0	1	0	1
1	0	1	1
1	0	0	1
1	0	1	0

Input: the row-by-row entities of the 4 x 4 matrix (i.e., 16 values of 0s or 1s). Each value is separated by a space.

Output: print “yes” if there exists a sequence of moves to flip the matrix into a zero matrix. Print “no” otherwise.

You need to compute the answer via a program (not just printing the final answer you computed somewhere else).